THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

DATE: Dec 20, 2023 TIME: 9:30am - 12:30pm

Answer ALL questions.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

QUESTION 1 (28 marks)

(a) **(14 marks)** The design and build (D&B) procurement system is becoming more favoured nowadays in major works projects as opposed to the traditional procurement system. Describe the engagement process and the contractual relationship adopted in the D&B and traditional system and discuss the merits of the D&B system over the traditional system that could explain the increase in popularity of the D&B system.

(b) **(7 marks)** A client wishes to adopt the management procurement system for a construction project but is concerned about the potential insolvency of the management contractor as the project progresses and the detrimental consequences to the project due to the insolvency. Please advise the client the variation of management procurement system that he should adopt in view of his concerns and provide reason for your advice.

(c) **(7 marks)** Discuss why the Build-Own-Operate-Transfer system is an attractive procurement system to be adopted for major road tunnel construction projects.

QUESTION 2 (22 marks)

The following table shows the immediately preceding activity(ies) (IPA), logical relationship, and duration of the activities involved in a construction project.

Activity ID	IPA	Duration (Days)
A	-	5
B	A	10
C	B	15
D	C	20
E	C(SS5)	10
F	C	10
G	D, F	12
H	E(SS2), G	20
I	G(FF3)	25
J	I	15
K	H, J	10

In the table above,
SS# = start-to-start relationship with a lag of #
FF# = finish-to-finish relationship with a lag of #

(a) **(13 marks)** Construct an activity-on-node network diagram based on the information provided in the table and identify the critical path(s) for this project.

(b) **(4 marks)** Explain the meaning of free float and determine the total and free floats for the non-critical activities.

(c) **(5 marks)** Describe what measures should be taken by the project manager to monitor and control project progress.

QUESTION 3 (28 marks)

A client is presented with four investment options, namely Option A, B, C, and D, with the projected revenue and costs provided in the table below.

OPTION A		
Investment period	4 years	
Costs	Capital cost	\$600k initially
	Management fee	\$80k at the end of each year for the entire investment period
Revenue	Investment return	\$280k at the end of each year for the entire investment period

OPTION B		
Investment period	5 years	
Costs	Capital cost	\$430k initially
	Management fee	\$20k at the end of each year for the entire investment period
Revenue	Investment return	\$100k at the end of Year 1; and \$300k at the end of Year 2 and 5

OPTION C		
Investment period	3 years	
Costs	Capital cost	\$720k initially; and \$50k at the end of Year 1
	Management fee	\$50k at the end of Year 1 and 2; and \$30k at the end of Year 3
Revenue	Investment return	\$350k at the end of each year for the entire investment period

OPTION D		
Investment period	5 years	
Costs	Capital cost	\$300k initially; \$50k at the end of Year 1; and \$50k at the end of Year 2
	Management fee	\$30k at the end of Year 1 and 3; and \$50k at the end of Year 5
Revenue	Investment return	\$130k at the end of Year 1, 2 and 3; and \$160k at the end of Year 4 and 5

Please answer the following questions. Table of values and formulae for compounding / discounting factors are provided in the Appendix for your use.

(a) **(15 marks)** If the discount rate is to be 8%, which option is the best for the client based on an evaluation using the incremental B/C ratio?

(b) **(6 marks)** If the client would like a rate of return of at least 5%, should he invest in Option C based on an evaluation using the internal rate or return (IRR)?

(c) **(3 marks)** Is it worthwhile for the client to invest in Option A if this option is to be evaluated by the net annual benefit with a discount rate of 10%?

(d) **(4 marks)** Explain the concept of time value of money.

QUESTION 4 (22 marks)

(a) **(6 marks)** The government is putting forward a major construction project with a project sum exceeding $10 billion. The main works in this project are technically complex and specialized. Which competitive tendering approach should the government select? Discuss the mechanism behind the selected tendering approach and explain why the government should adopt this approach.

(b) **(10 marks)** Explain what a functional organization structure is and discuss the pros and cons for an organization to adopt such structure. Provide an example of functional organizational structure for a civil engineering consulting firm.

(c) **(6 marks)** Explain the purpose for carrying out feasibility study in a construction project and discuss the benefits to the general public from carrying out a feasibility study.

$i = 5\%$

n	caf $(1+i)^n$	pwf $\frac{1}{(1+i)^n}$	uscaf $\frac{(1+i)^n-1}{i}$	ussff $\frac{i}{(1+i)^n-1}$	uspwf $\frac{(1+i)^n-1}{i(1+i)^n}$	uscrf $\frac{i(1+i)^n}{(1+i)^n-1}$
1	1.0500	0.9524	1.0000	1.0000	0.9524	1.0500
2	1.1025	0.9070	2.0500	0.4878	1.8594	0.5378
3	1.1576	0.8638	3.1525	0.3172	2.7232	0.3672
4	1.2155	0.8227	4.3101	0.2320	3.5460	0.2820
5	1.2763	0.7835	5.5256	0.1810	4.3295	0.2310
6	1.3401	0.7462	6.8019	0.1470	5.0757	0.1970
7	1.4071	0.7107	8.1420	0.1228	5.7864	0.1728
8	1.4775	0.6768	9.5491	0.1047	6.4632	0.1547
9	1.5513	0.6446	11.0266	0.0907	7.1078	0.1407
10	1.6289	0.6139	12.578	0.0795	7.7217	0.1295
11	1.7103	0.5847	14.207	0.0704	8.3064	0.1204
12	1.7959	0.5568	15.917	0.0628	8.863	0.1128
13	1.8856	0.5303	17.713	0.0565	9.394	0.1065
14	1.9799	0.5051	19.599	0.0510	9.899	0.1010
15	2.0789	0.4810	21.579	0.0463	10.380	0.0963
16	2.1829	0.4581	23.657	0.0423	10.838	0.0923
17	2.2920	0.4363	25.840	0.0387	11.274	0.0887
18	2.4066	0.4155	28.132	0.0355	11.690	0.0855
19	2.5270	0.3957	30.539	0.0327	12.085	0.0827
20	2.6533	0.3769	33.066	0.0302	12.462	0.0802
21	2.7860	0.3589	35.719	0.0280	12.821	0.0780
22	2.9253	0.3418	38.505	0.0260	13.163	0.0760
23	3.0715	0.3256	41.430	0.0241	13.489	0.0741
24	3.2251	0.3101	44.502	0.0225	13.799	0.0725
25	3.3864	0.2953	47.727	0.0210	14.094	0.0710
26	3.5557	0.2812	51.113	0.0196	14.375	0.0696
27	3.7335	0.2678	54.669	0.0183	14.643	0.0683
28	3.9201	0.2551	58.403	0.0171	14.898	0.0671
29	4.1161	0.2429	62.323	0.0160	15.141	0.0660
30	4.3219	0.2314	66.439	0.0151	15.372	0.0651

$i = 6\%$

n	caf $(1+i)^n$	pwf $\frac{1}{(1+i)^n}$	uscaf $\frac{(1+i)^n-1}{i}$	ussff $\frac{i}{(1+i)^n-1}$	uspwf $\frac{(1+i)^n-1}{i(1+i)^n}$	uscrf $\frac{i(1+i)^n}{(1+i)^n-1}$
1	1.0600	0.9434	1.0000	1.0000	0.9434	1.0600
2	1.1236	0.8900	2.0600	0.4854	1.8334	0.5454
3	1.1910	0.8396	3.1836	0.3141	2.6730	0.3741
4	1.2625	0.7921	4.3746	0.2286	3.4651	0.2886
5	1.3382	0.7473	5.6371	0.1774	4.2124	0.2374
6	1.4185	0.7050	6.9753	0.1434	4.9173	0.2034
7	1.5036	0.6651	8.3938	0.1191	5.5824	0.1791
8	1.5938	0.6274	9.8975	0.1010	6.2098	0.1610
9	1.6895	0.5919	11.4913	0.0870	6.8017	0.1470
10	1.7908	0.5584	13.181	0.0759	7.3601	0.1359
11	1.8983	0.5268	14.972	0.0668	7.8869	0.1268
12	2.0122	0.4970	16.870	0.0593	8.384	0.1193
13	2.1329	0.4688	18.882	0.0530	8.853	0.1130
14	2.2609	0.4423	21.015	0.0476	9.295	0.1076
15	2.3966	0.4173	23.276	0.0430	9.712	0.1030
16	2.5404	0.3936	25.673	0.0390	10.106	0.0990
17	2.6928	0.3714	28.213	0.0354	10.477	0.0954
18	2.8543	0.3503	30.906	0.0324	10.828	0.0924
19	3.0256	0.3305	33.760	0.0296	11.158	0.0896
20	3.2071	0.3118	36.786	0.0272	11.470	0.0872
21	3.3996	0.2942	39.993	0.0250	11.764	0.0850
22	3.6035	0.2775	43.392	0.0230	12.042	0.0830
23	3.8197	0.2618	46.996	0.0213	12.303	0.0813
24	4.0489	0.2470	50.816	0.0197	12.550	0.0797
25	4.2919	0.2330	54.865	0.0182	12.783	0.0782
26	4.5494	0.2198	59.156	0.0169	13.003	0.0769
27	4.8223	0.2074	63.706	0.0157	13.211	0.0757
28	5.1117	0.1956	68.528	0.0146	13.406	0.0746
29	5.4184	0.1846	73.640	0.0136	13.591	0.0736
30	5.7435	0.1741	79.058	0.0126	13.765	0.0726

$i = 7\%$

n	caf $(1+i)^n$	pwf $\frac{1}{(1+i)^n}$	uscaf $\frac{(1+i)^n-1}{i}$	ussff $\frac{i}{(1+i)^n-1}$	uspwf $\frac{(1+i)^n-1}{i(1+i)^n}$	uscrf $\frac{i(1+i)^n}{(1+i)^n-1}$
1	1.0700	0.9346	1.0000	1.0000	0.9346	1.0700
2	1.1449	0.8734	2.0700	0.4831	1.8080	0.5531
3	1.2250	0.8163	3.2149	0.3111	2.6243	0.3811
4	1.3108	0.7629	4.4399	0.2252	3.3872	0.2952
5	1.4026	0.7130	5.7507	0.1739	4.1002	0.2439
6	1.5007	0.6663	7.1533	0.1398	4.7665	0.2098
7	1.6058	0.6227	8.6540	0.1156	5.3893	0.1856
8	1.7182	0.5820	10.2598	0.0975	5.9713	0.1675
9	1.8385	0.5439	11.9780	0.0835	6.5152	0.1535
10	1.9672	0.5083	13.816	0.0724	7.0236	0.1424
11	2.1049	0.4751	15.784	0.0634	7.4987	0.1334
12	2.2522	0.4440	17.888	0.0559	7.943	0.1259
13	2.4098	0.4150	20.141	0.0497	8.358	0.1197
14	2.5785	0.3878	22.550	0.0443	8.745	0.1143
15	2.7590	0.3624	25.129	0.0398	9.108	0.1098
16	2.9522	0.3387	27.888	0.0359	9.447	0.1059
17	3.1588	0.3166	30.840	0.0324	9.763	0.1024
18	3.3799	0.2959	33.999	0.0294	10.059	0.0994
19	3.6165	0.2765	37.379	0.0268	10.336	0.0968
20	3.8697	0.2584	40.995	0.0244	10.594	0.0944
21	4.1406	0.2415	44.865	0.0223	10.836	0.0923
22	4.4304	0.2257	49.006	0.0204	11.061	0.0904
23	4.7405	0.2109	53.436	0.0187	11.272	0.0887
24	5.0724	0.1971	58.177	0.0172	11.469	0.0872
25	5.4274	0.1842	63.249	0.0158	11.654	0.0858
26	5.8074	0.1722	68.676	0.0146	11.826	0.0846
27	6.2139	0.1609	74.484	0.0134	11.987	0.0834
28	6.6488	0.1504	80.698	0.0124	12.137	0.0824
29	7.1143	0.1406	87.347	0.0114	12.278	0.0814
30	7.6123	0.1314	94.461	0.0106	12.409	0.0806

$i = 8\%$

n	caf $(1+i)^n$	pwf $\frac{1}{(1+i)^n}$	uscaf $\frac{(1+i)^n-1}{i}$	ussff $\frac{i}{(1+i)^n-1}$	uspwf $\frac{(1+i)^n-1}{i(1+i)^n}$	uscrf $\frac{i(1+i)^n}{(1+i)^n-1}$
1	1.0800	0.9259	1.0000	1.0000	0.9259	1.0800
2	1.1664	0.8573	2.0800	0.4808	1.7833	0.5608
3	1.2597	0.7938	3.2464	0.3080	2.5771	0.3880
4	1.3605	0.7350	4.5061	0.2219	3.3121	0.3019
5	1.4693	0.6806	5.8666	0.1705	3.9927	0.2505
6	1.5869	0.6302	7.3359	0.1363	4.6229	0.2163
7	1.7138	0.5835	8.9228	0.1121	5.2064	0.1921
8	1.8509	0.5403	10.6366	0.0940	5.7466	0.1740
9	1.9990	0.5002	12.4876	0.0801	6.2469	0.1601
10	2.1589	0.4632	14.487	0.0690	6.7101	0.1490
11	2.3316	0.4289	16.645	0.0601	7.1390	0.1401
12	2.5182	0.3971	18.977	0.0527	7.536	0.1327
13	2.7196	0.3677	21.495	0.0465	7.904	0.1265
14	2.9372	0.3405	24.215	0.0413	8.244	0.1213
15	3.1722	0.3152	27.152	0.0368	8.559	0.1168
16	3.4259	0.2919	30.324	0.0330	8.851	0.1130
17	3.7000	0.2703	33.750	0.0296	9.122	0.1096
18	3.9960	0.2502	37.450	0.0267	9.372	0.1067
19	4.3157	0.2317	41.446	0.0241	9.604	0.1041
20	4.6610	0.2145	45.762	0.0219	9.818	0.1019
21	5.0338	0.1987	50.423	0.0198	10.017	0.0998
22	5.4365	0.1839	55.457	0.0180	10.201	0.0980
23	5.8715	0.1703	60.893	0.0164	10.371	0.0964
24	6.3412	0.1577	66.765	0.0150	10.529	0.0950
25	6.8485	0.1460	73.106	0.0137	10.675	0.0937
26	7.3964	0.1352	79.954	0.0125	10.810	0.0925
27	7.9881	0.1252	87.351	0.0114	10.935	0.0914
28	8.6271	0.1159	95.339	0.0105	11.051	0.0905
29	9.3173	0.1073	103.966	0.0096	11.158	0.0896
30	10.0627	0.0994	113.283	0.0088	11.258	0.0888

$i = 9\%$

n	caf $(1+i)^n$	pwf $\frac{1}{(1+i)^n}$	uscaf $\frac{(1+i)^n-1}{i}$	ussff $\frac{i}{(1+i)^n-1}$	uspwf $\frac{(1+i)^n-1}{i(1+i)^n}$	uscrf $\frac{i(1+i)^n}{(1+i)^n-1}$
1	1.0900	0.9174	1.0000	1.0000	0.9174	1.0900
2	1.1881	0.8417	2.0900	0.4785	1.7591	0.5685
3	1.2950	0.7722	3.2781	0.3051	2.5313	0.3951
4	1.4116	0.7084	4.5731	0.2187	3.2397	0.3087
5	1.5386	0.6499	5.9847	0.1671	3.8897	0.2571
6	1.6771	0.5963	7.5233	0.1329	4.4859	0.2229
7	1.8280	0.5470	9.2004	0.1087	5.0330	0.1987
8	1.9926	0.5019	11.0285	0.0907	5.5348	0.1807
9	2.1719	0.4604	13.0210	0.0768	5.9952	0.1668
10	2.3674	0.4224	15.193	0.0658	6.4177	0.1558
11	2.5804	0.3875	17.560	0.0569	6.8052	0.1469
12	2.8127	0.3555	20.141	0.0497	7.161	0.1397
13	3.0658	0.3262	22.953	0.0436	7.487	0.1336
14	3.3417	0.2992	26.019	0.0384	7.786	0.1284
15	3.6425	0.2745	29.361	0.0341	8.061	0.1241
16	3.9703	0.2519	33.003	0.0303	8.313	0.1203
17	4.3276	0.2311	36.974	0.0270	8.544	0.1170
18	4.7171	0.2120	41.301	0.0242	8.756	0.1142
19	5.1417	0.1945	46.018	0.0217	8.950	0.1117
20	5.6044	0.1784	51.160	0.0195	9.129	0.1095
21	6.1088	0.1637	56.765	0.0176	9.292	0.1076
22	6.6586	0.1502	62.873	0.0159	9.442	0.1059
23	7.2579	0.1378	69.532	0.0144	9.580	0.1044
24	7.9111	0.1264	76.790	0.0130	9.707	0.1030
25	8.6231	0.1160	84.701	0.0118	9.823	0.1018
26	9.3992	0.1064	93.324	0.0107	9.929	0.1007
27	10.2451	0.0976	102.723	0.0097	10.027	0.0997
28	11.1671	0.0895	112.968	0.0089	10.116	0.0989
29	12.1722	0.0822	124.135	0.0081	10.198	0.0981
30	13.2677	0.0754	136.308	0.0073	10.274	0.0973

$i = 10\%$

n	caf $(1+i)^n$	pwf $\frac{1}{(1+i)^n}$	uscaf $\frac{(1+i)^n-1}{i}$	ussff $\frac{i}{(1+i)^n-1}$	uspwf $\frac{(1+i)^n-1}{i(1+i)^n}$	uscrf $\frac{i(1+i)^n}{(1+i)^n-1}$
1	1.1000	0.9091	1.0000	1.0000	0.9091	1.1000
2	1.2100	0.8264	2.1000	0.4762	1.7355	0.5762
3	1.3310	0.7513	3.3100	0.3021	2.4869	0.4021
4	1.4641	0.6830	4.6410	0.2155	3.1699	0.3155
5	1.6105	0.6209	6.1051	0.1638	3.7908	0.2638
6	1.7716	0.5645	7.7156	0.1296	4.3553	0.2296
7	1.9487	0.5132	9.4872	0.1054	4.8684	0.2054
8	2.1436	0.4665	11.4359	0.0874	5.3349	0.1874
9	2.3579	0.4241	13.5795	0.0736	5.7590	0.1736
10	2.5937	0.3855	15.937	0.0627	6.1446	0.1627
11	2.8531	0.3505	18.531	0.0540	6.4951	0.1540
12	3.1384	0.3186	21.384	0.0468	6.814	0.1468
13	3.4523	0.2897	24.523	0.0408	7.103	0.1408
14	3.7975	0.2633	27.975	0.0357	7.367	0.1357
15	4.1772	0.2394	31.772	0.0315	7.606	0.1315
16	4.5950	0.2176	35.950	0.0278	7.824	0.1278
17	5.0545	0.1978	40.545	0.0247	8.022	0.1247
18	5.5599	0.1799	45.599	0.0219	8.201	0.1219
19	6.1159	0.1635	51.159	0.0195	8.365	0.1195
20	6.7275	0.1486	57.275	0.0175	8.514	0.1175
21	7.4002	0.1351	64.002	0.0156	8.649	0.1156
22	8.1403	0.1228	71.403	0.0140	8.772	0.1140
23	8.9543	0.1117	79.543	0.0126	8.883	0.1126
24	9.8497	0.1015	88.497	0.0113	8.985	0.1113
25	10.8347	0.0923	98.347	0.0102	9.077	0.1102
26	11.9182	0.0839	109.182	0.0092	9.161	0.1092
27	13.1100	0.0763	121.100	0.0083	9.237	0.1083
28	14.4210	0.0693	134.210	0.0075	9.307	0.1075
29	15.8631	0.0630	148.631	0.0067	9.370	0.1067
30	17.4494	0.0573	164.494	0.0061	9.427	0.1061

APPENDIX (CONTINUED)

Compound / Discounting Formula

Compound amount factor	$\text{caf} = (1+i)^n$
Uniform series compound amount factor	$\text{uscaf} = \frac{(1+i)^n-1}{i}$
Uniform series sinking found factor	$\text{ussff} = \frac{i}{(1+i)^n-1}$
Uniform series capital recovery factor	$\text{uscrf} = \frac{i(1+i)^n}{(1+i)^n-1}$
Present worth factor	$\text{pwf} = \frac{1}{(1+i)^n}$
Uniform series present worth factor	$\text{uspwf} = \frac{(1+i)^n-1}{i(1+i)^n}$

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